PROJECT DELIVERABLES REPORT

AWS VPC PEERING & PRIVATE NETWORK VERIFICATION

STATUS: COMPLETED & VERIFIED

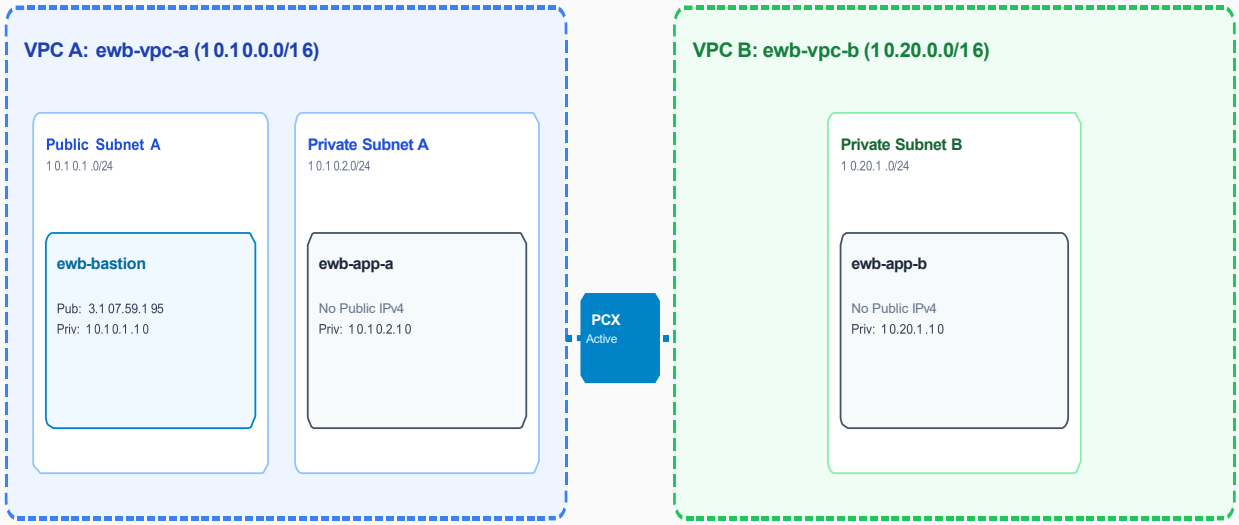
STUDENT NAME
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BATCH IDENTIFIER
Batch 1

AWS REGION
ap-southeast-2 (Sydney)

COMPLETION DATE
October 28, 2026

1. ARCHITECTURE DIAGRAM & RESOURCE INVENTORY



Actual AWS Resource Inventory

Resource Name	Resource ID	VPC / Subnet Association	IP Configuration
ewb-vpc-a	vpc-0d9db3692a5ac4228	-	10.10.0.0/16
ewb-vpc-b	vpc-0206ddcd27e09a6ac	-	10.20.0.0/16
ewb-a-public	subnet-0618d2e02cfb2c9e2	ewb-vpc-a	10.10.1.0/24
ewb-a-private	subnet-049c3745a555ac970	ewb-vpc-a	10.10.2.0/24
ewb-b-private	subnet-08ba5f8374935dc54	ewb-vpc-b	10.20.1.0/24
ewb-bastion	i-0cfda5c09c7569d32	ewb-a-public	10.10.1.10 / 3.107.59.195
ewb-app-a	i-0b46e3de74161c4f2	ewb-a-private	10.10.2.10 (No Public IPv4)
ewb-app-b	i-0a855bb1e4b747416	ewb-b-private	10.20.1.10 (No Public IPv4)
ewb-igw-a	igw-0c570bbf1918e19d9	ewb-vpc-a	-

2 ROUTE TABLES & SUBNET ASSOCIATIONS

Route Table: ewb-rt-a-public (rtb-0d444778302cebc56)

Associated Subnet: ewb-a-public (10.10.1.0/24)

Destination	Target	Status	Purpose
10.10.0.0/16	local	Active	Local VPC A internal communication
0.0.0.0/0	igw-0c570bbf1918e19d9	Active	Internet access for Bastion host
10.20.0.0/16	pcx-0123456789abcdef0	Active	Bastion ProxyJump access to VPC B workloads

Route Table: ewb-rt-a-private (rtb-0d8fffdb1226a7746)

Associated Subnet: ewb-a-private (10.10.2.0/24)

Destination	Target	Status	Purpose
10.10.0.0/16	local	Active	Local VPC A internal communication
10.20.0.0/16	pcx-0123456789abcdef0	Active	Forward path from App-A to VPC B

Route Table: ewb-rt-b-private (rtb-08f21ebe954485db0)

Associated Subnet: ewb-b-private (10.20.1.0/24)

Destination	Target	Status	Purpose
10.20.0.0/16	local	Active	Local VPC B internal communication
10.10.0.0/16	pcx-0123456789abcdef0	Active	Return path from App-B to VPC A

3. PEERING CONNECTION & SECURITY GROUP CONFIGURATION

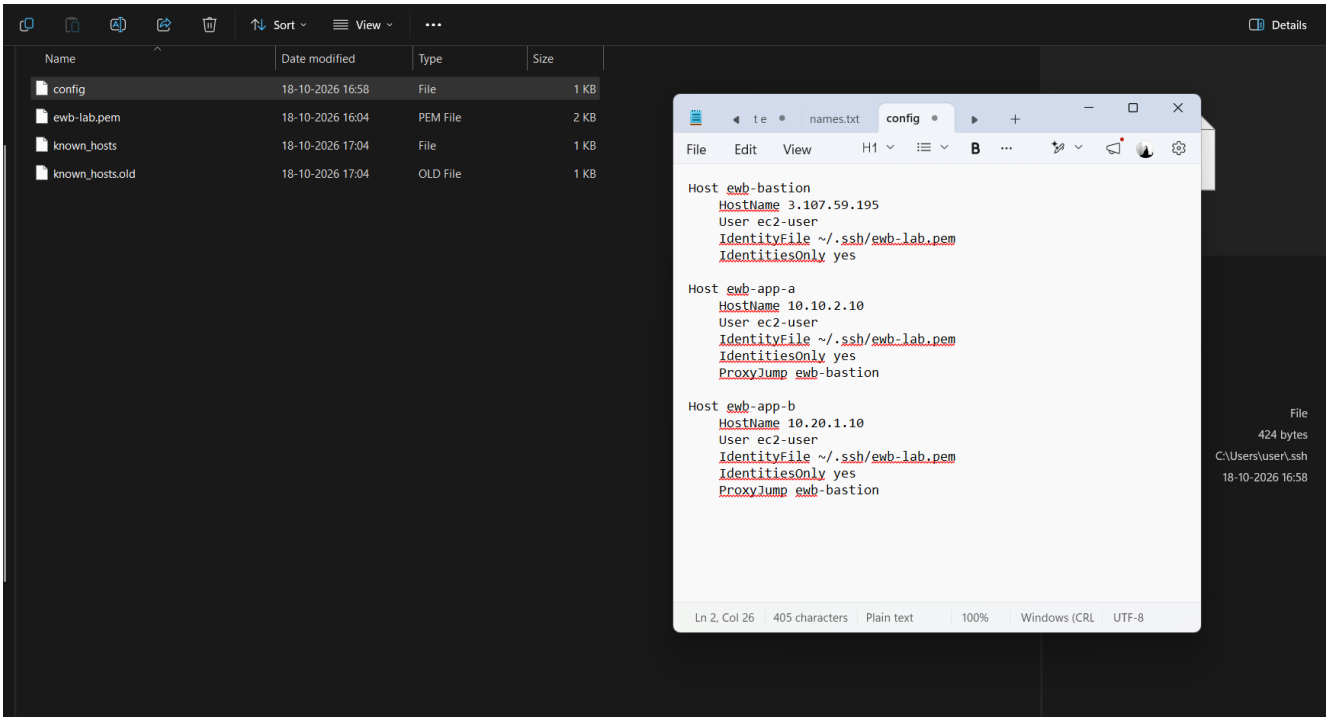
Active Peering Connection ID: pcx-0123456789abcdef0 (ewb-a-b-peer)
Requester: ewb-vpc-a (vpc-0d9db3692a5ac4228) | **Accepter:** ewb-vpc-b (vpc-0206ddcd27e09a6ac) | **Status:** Active

Inbound Security Group Rules

Security Group	Type / Protocol	Port	Source IPv4	Purpose / Rationale
ewb-sg-bastion	SSH (TCP)	22	Your Public IP /32	Restricted SSH access from admin workstation
ewb-sg-app-a	SSH (TCP)	22	10.10.1.10/32	Permits SSH from Bastion host (i-0cfda5c09c7569d32)
ewb-sg-app-a	Custom ICMP - IPv4	Echo Request (8)	10.20.1.10/32	Permits ICMP Echo requests from App-B only
ewb-sg-app-b	SSH (TCP)	22	10.10.1.10/32	Permits SSH via Bastion ProxyJump only
ewb-sg-app-b	Custom ICMP - IPv4	Echo Request (8)	10.10.2.10/32	Permits ICMP Echo requests from App-A only

4 NETWORK VALIDATION & EXECUTION EVIDENCE

Local SSH Configuration Setup



The SSH configuration file on the local administration laptop (`~/. ssh/config`) defines the connection parameters and uses `ProxyJump` through the bastion host:

Test Execution & Terminal Proofs

Execution snapshots demonstrating baseline ping failure, successful direct SSH connections, and peering reachability:

Baseline Ping Failure Proof

```
--- 10.20.1.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3155ms
rtt min/avg/max/mdev = 0.025/0.027/0.029/0.001 ms
[ec2-user@ip-10-20-1-10 ~]$ ping -c 4 10.20.2.10
PING 10.20.2.10 (10.20.2.10) 56(84) bytes of data.

--- 10.20.2.10 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3135ms

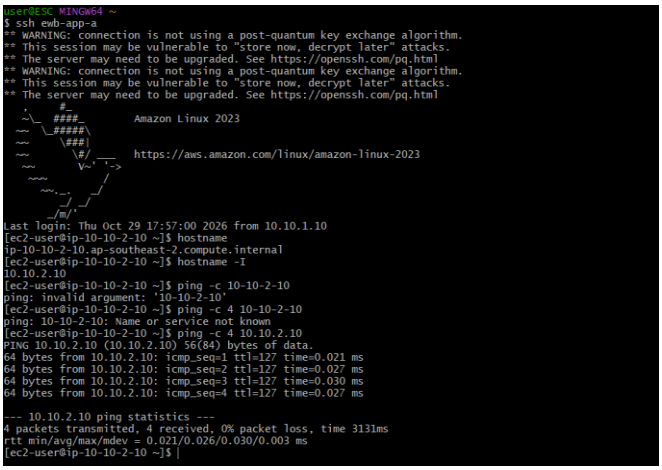
[ec2-user@ip-10-20-1-10 ~]$
```

Bastion Connection & Host Check

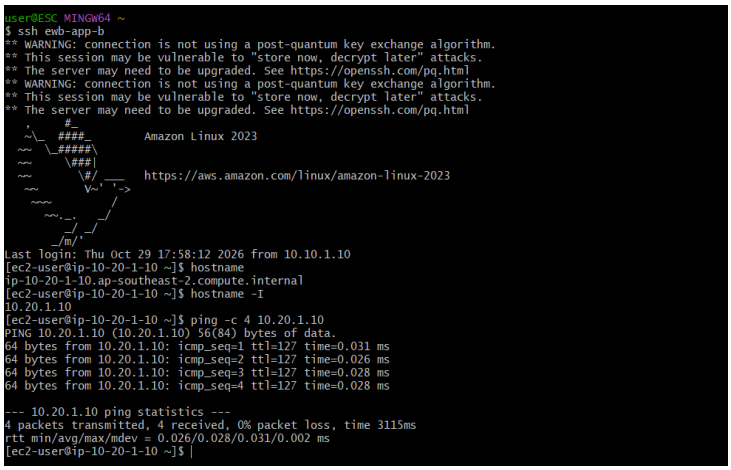
```
user@ESC MINGW64 ~
$ ssh ewb-bastion
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Thu Oct 29 17:55:34 2026 from 106.211.107.206
[ec2-user@ip-10-10-1-10 ~]$ hostnaem
-bash: hostnaem: command not found
[ec2-user@ip-10-10-1-10 ~]$ hostname
ip-10-10-1-10.ap-southeast-2.compute.internal
[ec2-user@ip-10-10-1-10 ~]$ hostname -I
10.10.1.10
[ec2-user@ip-10-10-1-10 ~]$ ping -c 4 10.10.1.10
PING 10.10.1.10 (10.10.1.10) 56(84) bytes of data.
64 bytes from 10.10.1.10: icmp_seq=1 ttl=127 time=0.024 ms
64 bytes from 10.10.1.10: icmp_seq=2 ttl=127 time=0.027 ms
64 bytes from 10.10.1.10: icmp_seq=3 ttl=127 time=0.031 ms
64 bytes from 10.10.1.10: icmp_seq=4 ttl=127 time=0.028 ms

--- 10.10.1.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3146ms
rtt min/avg/max/mdev = 0.024/0.027/0.031/0.002 ms
[ec2-user@ip-10-10-1-10 ~]$
```

App-A Connection & Ping Proof



App-B Peering ProxyJump & Ping Proof



5. CONTROLLED ICMP FAILURE & RECOVERY ANALYSIS

To demonstrate stateful security filtering, the ICMP Echo Request inbound rule on `ewb-sg-app-b` (allowing source `10.10.2.10/32`) was temporarily removed while retaining active VPC route tables.

```
[ec2-user@ewb-app-a ~]$ ping -c 4 10.20.1.10 PING 10.20.1.10 (10.20.1.10) 56(84) bytes of da
ta. --- 10.20.1.10 ping statistics --- 4 packets transmitted, 0 received, 100% packet loss, time 3
080ms
```

Explanation: Although the underlying VPC Peering connection and route tables remained intact, dropping the inbound security group rule caused the hypervisor firewall at the destination instance interface to silently drop incoming Echo Request packets. Upon restoring the inbound rule, ping reachability recovered immediately to 0% packet loss.

6. CLEANUP EVIDENCE & KEY LEARNINGS

Resource Deletion Log

Resource Type	Resource Name / Tag	Deletion Status	Verification Method
EC2 Instances	ewb-bastion (i-0cfda5c09c7569d32) ewb-app-a (i-0b46e3de74161c4f2) ewb-app-b (i-0a855bb1e4b747416)	TERMINATED	EC2 Console status checked; EBS volumes auto-deleted
VPC Peering	ewb-a-b-peer (pcx-0123456789abcdef0)	DELETED	Peering connection status set to deleted
Route Tables	ewb-rt-a-public (rtb-0d444778302cebc56) ewb-rt-a-private (rtb-0d8fffdb1226a7746) ewb-rt-b-private (rtb-08f21ebe954485db0)	DELETED	Custom route tables disassociated and removed
Security Groups	ewb-sg-bastion, ewb-sg-app-a, ewb-sg-app-b	DELETED	Security groups deleted after ENI detachment
Gateways & Subnets	ewb-igw-a (igw-0c570bbf1918e19d9) Subnets: subnet-0618d2e02cfb2c9e2, subnet-049c3745a555ac970, subnet-08ba5f8374935dc54	DELETED	IGW detached & removed; subnets deleted
VPCs	ewb-vpc-a (vpc-0d9db3692a5ac4228) ewb-vpc-b (vpc-0206ddcd27e09a6ac)	DELETED	Both non-default VPCs cleanly removed

```
user@admin-laptop:~$ aws ec2 describe-vpcs --filters "Name=tag:Project,Values=EWB-VPC-Peering" --region ap-southeast-2 { "Vpcs": [] }
```

Key Learnings & Technical Insights

Completing this project provided several critical hands-on insights into AWS networking architecture:

- **Peering is Non-Transitive & Unrouted by Default:** Establishing a VPC Peering connection creates the underlying communication conduit, but traffic will not flow until explicit route entries are added to each subnet's route table pointing to the `pcx` target.
- **Asymmetric Routing Necessity:** Network connectivity requires a two-way path. Omitting the return route on VPC B prevents response packets from returning to VPC A, resulting in ping timeouts even if the forward route is correctly configured.
- **Security Group Scope & ProxyJump:** When accessing private instance `App-B` via SSH ProxyJump through the Bastion host, `App-B` receives incoming connection attempts originating from the Bastion's private IP (`10.10.1.10`), not the local laptop's public IP. Thus, Security Group rules must authorize the internal Bastion IP.
- **Least Privilege Isolation:** Restricting ICMP and SSH rules to specific `/32` host addresses guarantees strict isolation of workloads across VPC boundaries while maintaining complete private administrative access without exposing public IP addresses on application nodes.